

Flow 5: Transaction Monitoring

Product Area: Bank / Program Admin

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Status: Draft

1. Purpose & Scope

This PRD defines the requirements for **Transaction Monitoring** in the Bank / Program Admin console — a real-time operational surface for viewing, filtering, investigating, and acting on cardholder transactions, declines, and fraud events.

This document covers:

- Real-time transaction feed (authorizations, posts, declines)
- Transaction search and filtering
- Pre-authorization hold and subscription recognition
- Decline metadata visibility
- Fraud alert review and action (confirm fraud, dismiss, freeze card)
- Cross-link to disputes and account management

This document does not define:

- Spend control configuration (Flow 7) — this surface *displays the effect of* spend control declines but does not edit rules
 - Statement-level aggregation (Flow 6)
 - Cardholder-facing transaction surfaces (End-Client PRDs)
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2. Goals & Success Criteria

Goal	Success Criterion
Operations can see and act on transaction events as they happen	Transaction feed shows new events within seconds of authorization

Goal	Success Criterion
Decline reasons are clear and human-readable	Every decline has a controlled-vocabulary reason code mapped to plain-English text
Fraud signals are surfaced and actionable	Flagged transactions are sortable, filterable, and have one-click escalation paths
Transaction context supports dispute and customer service workflows	A single transaction record links to merchant, account, dispute, and statement context



3. Personas

- **Operations / Servicing Agent (primary):** Looks up transactions to support customer calls and resolve issues.
- **Fraud Analyst:** Reviews flagged activity and decides on actions (freeze, dismiss, escalate).
- **Program Admin:** Monitors program-wide transaction trends and anomalies.
- **Cardholder (downstream):** Sees their own transactions in-app, populated from the same upstream data source.



4. Functional Requirements

4.1 Real-Time Transaction Feed

Continuous stream of all transaction events across the program.

Event types displayed

- Authorization (pending)
- Authorization decline
- Post (capture / clearing)
- Refund / Credit
- Reversal (auth reversed before clearing)
- Adjustment (post-clearing change)
- Pre-authorization hold (large estimated holds — hotels, car rentals, gas)

- Subscription / recurring (system-tagged based on merchant pattern)

Per-row data

- Timestamp (UTC and local cardholder time)
- Cardholder + card reference (masked PAN last 4)
- Merchant name (network-reported + cleansed display name)
- MCC (code + category label)
- Amount + currency
- Status (Pending / Posted / Declined / Refunded / Cancelled / Reversed)
- Fraud flag indicator (if any)
- Dispute indicator (if a dispute is open or resolved)
- Network reference ID

4.2 Search & Filter

Filters available:

- Date range (defaults to last 24h on landing)
- Status
- Amount range
- MCC / category
- Merchant name (partial match)
- Cardholder / account
- Card (specific card if account has multiple)
- Country / region (for foreign transactions)
- Fraud-flagged only
- Disputed only
- Recurring / subscription only

Search by:

- Network reference ID
- Account ID / cardholder name
- Merchant
- Amount (exact or fuzzy)

4.3 Transaction Detail View

For each transaction, a detail panel with:

Core

- All row data plus original authorization vs. clearing amount diff (if any)
- Entry mode (chip, contactless, e-commerce, manual entry, etc.)
- Terminal / merchant ID
- Currency conversion details if applicable
- Reward points / cashback earned on this transaction
- Associated statement cycle

Status timeline

- Auth → Pending → Posted → (Refunded / Reversed / Disputed) — with timestamps for each transition

Decline details (if declined)

- Decline reason code (controlled vocabulary)
- Plain-language explanation
- Triggering rule (e.g., MCC blocked, velocity limit, geo restriction, NSF, fraud score) — links back to the relevant Flow 7 spend control where applicable

Pre-auth hold details (if applicable)

- Estimated hold amount
- Estimated release date
- Final cleared amount (when known)
- Adjustment delta vs. original hold

Subscription detection (if applicable)

- Pattern source (merchant + amount + cadence)
- Confidence label (system-assigned)
- Past instances of this subscription on the same card

Cross-links

- Linked dispute case (if any) — link to Flow 3
- Statement that includes this transaction — link to Flow 6
- Cardholder account — link to Flow 2

4.4 Decline Reason Vocabulary

A controlled list, each mapped to:

- Internal reason code
- Plain-language label for admin view
- Plain-language label for cardholder view (per the cardholder app's Decline Reasons feature)

Examples:

- `MCC_BLOCKED` — "Merchant category not permitted"
- `VELOCITY_DAILY_LIMIT` — "Too many transactions today"
- `GEO_RESTRICTED` — "Transaction outside permitted region"
- `INSUFFICIENT_AVAILABLE_CREDIT` — "Not enough available credit"
- `CASH_ADVANCE_DISABLED` — "Cash advance not enabled on this card"
- `FRAUD_SCORE_THRESHOLD` — "Flagged by fraud system"
- `CARD_FROZEN` — "Card is currently frozen"
- `CARD_NOT_ACTIVATED` — "Card has not been activated"
- `EXPIRED_CARD` — "Card expired"
- `PIN_INCORRECT` — "PIN failed verification"

Mapping is configurable; new codes added through a controlled release.

4.5 Fraud Alert Review

When the fraud engine flags a transaction (either post-decline or post-auth-but-pre-post), the transaction appears in a Fraud Alerts queue.

Per-alert data

- Transaction
- Fraud signal(s) that fired (e.g., unusual geography, velocity, merchant risk, amount anomaly)
- Risk score
- Cardholder's response, if any (from the "Was this you?" in-app confirmation — see Flow 11 cardholder edge cases)

Admin actions

- **Confirm fraud** — flags transaction, opens a fraud case, freezes card with reason `Bank-initiated: Suspected fraud`, notifies cardholder, optionally initiates auto-reissue
- **Dismiss** — clears the flag, transaction proceeds normally
- **Escalate** — routes to senior fraud analyst with notes
- **Contact cardholder** — triggers an out-of-band verification ("Was this you?") through the configured channel

Audit reference (Findings #9, #10): admin-driven card freezes must be tagged as Bank-initiated; this distinction propagates to the cardholder-facing freeze screen.

4.6 Pre-Authorization Holds

Pre-auth holds (typical at hotels, rental cars, gas pumps) appear in the feed with:

- Initial hold amount
- Expected release timeline (per merchant category convention)
- Cleared amount when known
- Delta calculation

This data flows through to the cardholder-facing transaction display (per MVP Feature B.4 “Display Pre-Authorization Holds”).

4.7 Subscription / Recurring Transaction Tagging

Basic heuristic recognition (same merchant + similar amount + ~monthly cadence). System-assigned tags are:

- `RECURRING_LIKELY` — strong pattern
- `RECURRING_POSSIBLE` — weaker pattern
- `NOT_RECURRING` — explicitly not (one-off)

Tag surfaces both in admin and cardholder views (cardholder sees “Recurring Merchant” label).

4.8 Bulk Actions (limited MVP scope)

- Export current view to CSV
- Bulk-flag a set of related transactions for fraud review

Bulk card freezes, bulk reversals, etc., are out of MVP scope.

4.9 Audit Log

Every admin action (confirm fraud, dismiss alert, manual freeze) logged with actor, timestamp, transaction reference, and reason.



5. Business Rules

1. The transaction feed is read-only with respect to transaction state itself — admins do not edit amounts, dates, or merchants.
2. Reversals and adjustments come from the payments / network systems; admins do not author reversals manually in this surface (those flow through dispute outcomes in Flow 3 or merchant-initiated refunds).
3. A transaction with an open dispute remains visible and continues to display the dispute link until the dispute is closed; the underlying transaction status does not change because of dispute state.

4. Pre-auth holds that expire without clearing (typical 30 days) are auto-released and marked **Expired Auth** — distinct from the dispute-window “Expired” terminology.

> *Audit reference (Section 8 terminology): “Expired” must be disambiguated. In transaction context it means “authorization expired without capture.” In dispute context it means “outside the FCBA window.” Surfaces using these terms must use unambiguous labels.*

6. Dependencies

Depends On	Reason
Flow 2: Cardholder Account Management	Account / card context per transaction
Flow 3: Dispute Management	Disputed transactions cross-link here
Flow 6: Statement & Billing	Transactions roll up into statements
Flow 7: Spend Controls	Source of decline rules that this surface displays the effect of
Card network authorization stream	Real-time event source
Fraud detection engine	Source of fraud flags and scores
End-Client: Transactions list, Pre-Auth Holds, Recurring Tags, Fraud Alerts	Consumer surfaces of this data

7. Edge Cases

Case	Expected Behavior
Authorization approved but never cleared (merchant doesn't capture)	After expiration window, marked Expired Auth ; available credit restored
Same transaction has multiple auths (partial captures, e.g., hotel)	Each auth visible; relationship to parent reservation shown if available
Currency conversion fee on foreign transaction	Displayed inline; reconciled to fee schedule from Flow 1

Case	Expected Behavior
Refund posts before the original transaction has cleared	Both appear; status shows the unusual ordering; netting logic in payments service handles the balance impact
Network reports duplicate transaction (network glitch)	Surfaced as separate events with a <code>DUPLICATE_SUSPECTED</code> flag; ops can confirm dedup
Cardholder reports “Not me” via in-app, but transaction was actually theirs	Fraud case opens; agent reviews and dismisses; transaction remains posted; cardholder is notified of resolution
Merchant-initiated reversal mid-investigation of a fraud case	Reversal posts; fraud case auto-flags for review and potential closure
Transaction declined for <code>CARD_FROZEN</code> , but cardholder didn’t know the card was frozen	Decline reason links back to freeze event timeline so agent can see when/why card was frozen
Subscription pattern wrongly tagged	Cardholder can untag in-app (post-MVP); admin can correct in detail view

8. Compliance Notes

- **CFPB UDAP / Reg E:** Decline reasons and fraud actions must be communicated clearly and promptly to cardholders. The decline-reason vocabulary and the bank-initiated freeze tagging are designed to support this.
- **Network rules (Visa/Mastercard):** Pre-auth hold release timing follows network conventions; admin surface displays expectations but does not override.
- **FCBA:** Cross-link from a transaction to dispute initiation flow assumes the dispute window check is enforced (per Flow 3 §4.1).
- **PCI DSS:** Full PAN is never displayed in admin surfaces (masked to last 4); sensitive data access is logged.

9. Open Questions

- Real-time push of new feed events vs. polling at N-second intervals — implementation choice.

- Granular permissions for fraud-action vs. read-only viewing — MVP role model?
- Should subscription tagging be machine-learned or strictly rule-based at MVP? (MVP: rule-based.)
- Future: merchant cleansing service for cardholder-friendly merchant names.
- Future: 3DS challenge surfacing in the feed for e-commerce transactions.